

12.213

AI25BTECH110031
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Question(12.213) The set V of all polynomials of a real variable x of degree two or less and with real coefficients, constitutes a real linear vector space $V = \{c_0 + c_1 x + c_2 x^2 : c_0, c_1, c_2 \in \mathbb{R}\}$

(a) For $f(x) = a_0 + a_1 x + a_2 x^2 \in V$ and $g(x) = b_0 + b_1 x + b_2 x^2 \in V$, which one of the following constitutes an acceptable scalar product?

- (i) $\langle f, g \rangle = a_0^2 b_0 + a_1^2 b_1 + a_2^2 b_2$
- (ii) $\langle f, g \rangle = a_0^2 b_0^2 + a_1^2 b_1^2 + a_2^2 b_2^2$
- (iii) $\langle f, g \rangle = a_0 b_0 - a_1 b_1 + a_2 b_2$
- (iv) $\langle f, g \rangle = a_0 b_0 + \frac{a_1 b_1}{2} + \frac{a_2 b_2}{3}$

(b) Using the scalar product obtained in the above question, identify the subspace of V that is orthogonal to $(1 + x)$.

- (i) $\{f(x) : b(1 - x) + cx^2, b, c \in \mathbb{R}\}$
- (ii) $\{f(x) : b(1 - 2x) + cx^2, b, c \in \mathbb{R}\}$
- (iii) $\{f(x) : b + cx^2, b, c \in \mathbb{R}\}$
- (iv) $\{f(x) : bx + cx^2, b, c \in \mathbb{R}\}$

Solution:

Given,

$$V = \{a_0 + a_1 x + a_2 x^2 : a_i \in \mathbb{R}\} \quad (4.1)$$

is a real vector space. For

$$f(x) = a_0 + a_1 x + a_2 x^2, \quad g(x) = b_0 + b_1 x + b_2 x^2, \quad (4.2)$$

A scalar product (inner product) $\langle f, g \rangle$ must satisfy:

1. Bilinearity,
2. Symmetry: $\langle f, g \rangle = \langle g, f \rangle$,
3. Positive definiteness: $\langle f, f \rangle > 0$ for all nonzero f .

Let us test each option:

- (i) $\langle f, g \rangle = a_0^2 b_0 + a_1^2 b_1 + a_2^2 b_2$ Not linear in f .
- (ii) $\langle f, g \rangle = a_0^2 b_0^2 + a_1^2 b_1^2 + a_2^2 b_2^2$ Not bilinear.
- (iii) $\langle f, g \rangle = a_0 b_0 - a_1 b_1 + a_2 b_2$ Not positive definite (e.g., for $f(x) = x$).
- (iv) $\langle f, g \rangle = a_0 b_0 + \frac{a_1 b_1}{2} + \frac{a_2 b_2}{3}$ Bilinear, symmetric, and positive definite.

Hence, (iv) is the acceptable scalar product.

(4.4)

Represent each polynomial $p(x) = a_0 + a_1x + a_2x^2$ by the coefficient vector

$$\mathbf{a} = \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix}. \quad (4.5)$$

The scalar product can be written as

$$\langle f, g \rangle = \mathbf{a}^T D \mathbf{b}, \quad D = \text{diag}\left(1, \frac{1}{2}, \frac{1}{3}\right). \quad (4.6)$$

Then the orthogonality condition $\langle f, 1+x \rangle = 0$ becomes

$$\mathbf{a}^T D \mathbf{u} = 0, \quad \mathbf{u} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}. \quad (4.7)$$

Compute

$$D \mathbf{u} = \begin{pmatrix} 1 \\ \frac{1}{2} \\ 0 \end{pmatrix}, \quad \Rightarrow a_0 + \frac{a_1}{2} = 0. \quad (4.8)$$

$$\Rightarrow a_0 = \frac{-a_1}{2} \quad (4.9)$$

and a_2 is independent

Let, $b = a_0$ and $c = a_2$

$$\mathbf{a} = b \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix} + c \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad (4.10)$$

So the orthogonal subspace is 2-dimensional with basis vectors corresponding to polynomials

$$1 - 2x, \quad x^2 \quad (4.11)$$

Hence every polynomial orthogonal to $1+x$ can be written

$$b(1 - 2x) + cx^2 \quad (4.12)$$

Orthogonal Subspace to $(1 + x)$ under Weighted Inner Product

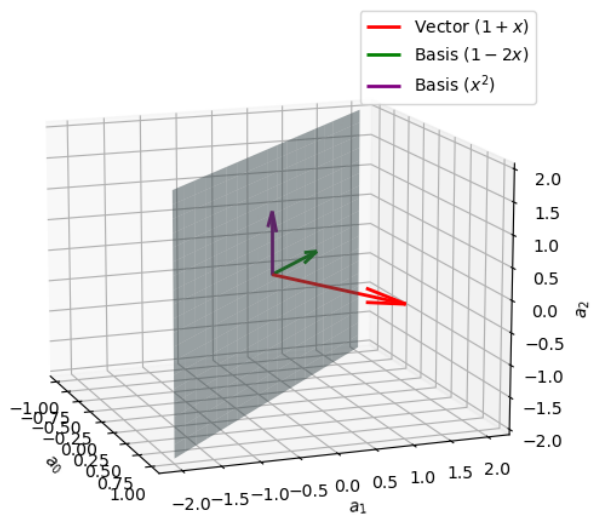


Fig. 4.1